KAIZEN MATHS

Lesson starter

Taylor and Maclaurin Series

Combining Series

Name

S1

Substitute carefully into the standard Maclaurin expansion.

Expand sin(2x) as a Maclaurin series up to and including the term in x^4.

S2

Multiply the relevant series and keep terms only up to the requested power of x.

Expand $e^2x \sin x$ up to and including the term in x^4.

S3

Use a standard reciprocal series or divide series to the requested order.

Find the Maclaurin series for 1 1 + 3x up to and including the term in x^4.

S4

Use the given series expansion to estimate the value.

Use a Maclaurin series to approximate ln(1.1).

Show your working

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Show your working

KAIZEN MATHS

Exit ticket

Taylor and Maclaurin Series

Combining Series

Name

EA1

Substitute carefully into the standard Maclaurin expansion.

Expand sin(2x) as a Maclaurin series up to and including the term in x^3.

EA2

Multiply the relevant series and keep terms only up to the requested power of x.

Expand $e^x \cos x$ up to and including the term in x^3.

Show your working

KAIZEN MATHS

Exit ticket

Taylor and Maclaurin Series

Combining Series

Name

EB1

Use a standard reciprocal series or divide series to the requested order.

Find the Maclaurin series for 1 1 - 2x up to and including the term in x^3.

Show your working

KAIZEN MATHS

Exit ticket

Taylor and Maclaurin Series

Combining Series

Name

EA1

Substitute carefully into the standard Maclaurin expansion.

Expand sin(2x) as a Maclaurin series up to and including the term in x^3.

EA2

Multiply the relevant series and keep terms only up to the requested power of x.

Expand $e^x \cos x$ up to and including the term in x^3.

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KAIZEN MATHS

Exit ticket

Taylor and Maclaurin Series

Combining Series

Name

EB1

Use a standard reciprocal series or divide series to the requested order.

Find the Maclaurin series for 1 1 - 2x up to and including the term in x^3.

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Teacher answers & working

Taylor and Maclaurin Series · Combining Series

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S1

Substitute carefully into the standard Maclaurin expansion.

Expand sin(2x) as a Maclaurin series up to and including the term in x^4.

Answer:

$2x - \frac{4}{3}x^3 + O(x^5)$

Working:

Start with the standard expansion: $\sin u = u - \frac{(u)^3}{3!} + \ldots.$

Substitute $u = 2x$.

After substitution: $\sin(2x) = 2x - \frac{(2x)^3}{3!} + \ldots.$

Simplify each substituted power and keep terms only up to x^4.

$\sin(2x) = 2x - \frac{4}{3}x^3 + O(x^5)$.

S2

Multiply the relevant series and keep terms only up to the requested power of x.

Expand $e^2x \sin x$ up to and including the term in x^4.

Answer:

$x + 2x^2 + \frac{11}{6}x^3 + x^4 + O(x^5)$

Working:

Write each factor as a series up to the order needed.

$e^{2x} = 1 + 2x + 2x^2 + \frac{4}{3}x^3 + \frac{2}{3}x^4 + O(x^5)$.

$\sin x = x - \frac{1}{6}x^3 + O(x^5)$.

Substitute these into the product: $e^{2x} \sin x = (1 + 2x + 2x^2 + \frac{4}{3}x^3 + \frac{2}{3}x^4 + O(x^5))(x - \frac{1}{6}x^3 + O(x^5))$.

Multiply term by term, collect like powers of x, and ignore terms above x^4.

$e^{2x} \sin x = x + 2x^2 + \frac{11}{6}x^3 + x^4 + O(x^5)$.

S3

Use a standard reciprocal series or divide series to the requested order.

Find the Maclaurin series for 1 1 + 3x up to and including the term in x^4.

Answer:

$1 - 3x + 9x^2 - 27x^3 + 81x^4 + O(x^5)$

Working:

Use $\frac{1}{1+u} = 1 - u + (u)^2 - (u)^3 + (u)^4 + \ldots.$

Here $u = 3x$.

After substitution: $\frac{1}{1+3x} = 1 - 3x + (3x)^2 - (3x)^3 + (3x)^4 + \ldots.$

Simplify to get $\frac{1}{1+3x} = 1 - 3x + 9x^2 - 27x^3 + 81x^4 + O(x^5)$.

Teacher answers & working

Taylor and Maclaurin Series · Combining Series

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S4

Use the given series expansion to estimate the value.

Use a Maclaurin series to approximate ln(1.1).

Answer:

0.09531

Working:

Use the expansion for ln(1 + x).

$\ln(1 + x) \approx x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4$.

Substitute x = 0.1.

$\ln(1.1) = \ln(1 + 0.1) \approx 0.1 - \frac{(0.1)^2}{2} + \frac{(0.1)^3}{3} - \frac{(0.1)^4}{4}$.

Calculate this numerical expression to get ln(1.1) ≈ 0.09531.

EA1

Substitute carefully into the standard Maclaurin expansion.

Expand sin(2x) as a Maclaurin series up to and including the term in x^3.

Answer:

$2x - \frac{4}{3}x^3 + O(x^4)$

Working:

Start with the standard expansion: $\sin u = u - \frac{(u)^3}{3!} + \ldots$.

Substitute u = 2x.

After substitution: $\sin(2x) = 2x - \frac{(2x)^3}{3!} + \ldots$.

Simplify each substituted power and keep terms only up to x^3.

$\sin(2x) = 2x - \frac{4}{3}x^3 + O(x^4)$.

EA2

Multiply the relevant series and keep terms only up to the requested power of x.

Expand $e^x \cos x$ up to and including the term in x^3.

Answer:

$1 + x - \frac{1}{3}x^3 + O(x^4)$

Working:

Write each factor as a series up to the order needed.

$e^x = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + O(x^4)$.

$\cos x = 1 - \frac{1}{2}x^2 + O(x^4)$.

Substitute these into the product: $e^x \cos x = (1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + O(x^4))(1 - \frac{1}{2}x^2 + O(x^4))$.

Multiply term by term, collect like powers of x, and ignore terms above x^3.

$e^x \cos x = 1 + x - \frac{1}{3}x^3 + O(x^4)$.

Teacher answers & working

Taylor and Maclaurin Series · Combining Series

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EB1

Use a standard reciprocal series or divide series to the requested order.

Find the Maclaurin series for 1 1 - 2x up to and including the term in x^3.

Answer:

$1 + 2x + 4x^2 + 8x^3 + O(x^4)$

Working:

Use $\frac{1}{1-u} = 1 + u + (u)^2 + (u)^3 + ...$

Here u = 2x.

After substitution: $\frac{1}{1-2x} = 1 + 2x + (2x)^2 + (2x)^3 + ...$

Simplify to get $\frac{1}{1-2x} = 1 + 2x + 4x^2 + 8x^3 + O(x^4)$.